

Roman Šafranko

Junior Software Engineer

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PROFILE

Computer science graduate focused on full-stack web development and applied machine learning. Experienced with React and Next.js, Python, PostgreSQL and Docker through self-directed projects taken from concept to deployment.

PROJECTS

Prismatic Arena 2026
2026

Teamfight Tactics statistics platform on Next.js and the Riot Games API, backed by PostgreSQL and Redis in Docker.
Designed an automated labelling pipeline clustering raw board states into named compositions, replacing hand-maintained tier lists.

VB Visualizer 2025

Browser music visualiser producing 9:16 video for short-form platforms, vanilla ES modules, Web Audio API, Canvas. No build step, no backend.
Implemented spectral-flux onset detection for beat-synced animation and LRCLIB integration for automatic lyric timing.

Voice-controlled robotic arm 2026

Speech to local LLM to motion planning on a Dobot Magician, with OpenCV closing the loop on object localisation.
Ran the full pipeline locally, speech recognition, inference and vision, with no cloud API.

nullres 2026

Purged walk-forward backtester
Walk-forward backtesting framework in Python/pandas for evaluating trading signals under strict leakage control.
Implemented purged walk-forward cross-validation, running each strategy with and without leakage to quantify lookahead-bias inflation.
Documented every disproved hypothesis with its metrics rather than only the surviving ones.

EDUCATION

B.Sc. Computer Science 2023 – 2026
Technical University of Košice · Košice
Thesis: Controlling the Dobot Magician robotic arm using multimodal language models.

SKILLS

JavaScript / TypeScript · React & Next.js · Node.js · Python · SQL & PostgreSQL · Redis · Linux · Computer vision · LLM integration

LANGUAGES

Slovak — Native English — C1